The K41 Spectrum as the Unique Variational Minimiser of a Scale-Resolved Free Energy

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Abstract

We formulate and prove a reference-normalised variational principle that characterises the Kolmogorov $k^{-5/3}$ energy spectrum as the unique energy-profile minimiser of a scale-resolved free-energy functional within the stated K41-normalised admissible class. Starting from a Littlewood–Paley decomposition of the velocity field, we define a free-energy functional $\mathcal{F}[\mathbf{E}]$ based on the Kullback–Leibler divergence between the shell-energy distribution and a K41 reference equilibrium. We introduce the K41-normalised class of admissible energy-flux operators—local, dimensionally consistent, monotone closures satisfying axioms (A1) and (A3) together with the normalisation condition in Definition 3.1—and prove that the K41 distribution is the unique minimiser of $\mathcal{F}[\mathbf{E}]$ over the joint feasible set of energy profiles and closures, subject to constant scale-to-scale energy flux (Theorem 3.5). The minimiser is non-degenerate: the Hessian of $\mathcal{F}$ at $\mathbf{E}^*$ is diagonal with entries $1/E_j^* > 0$, providing shell-energy stiffness that grows as $k_j^{2/3}$ for dyadic shells. We also record two guardrails: relaxed flux constraints do not yield a non-trivial robustness theorem unless the exact K41 profile is excluded or the objective is changed, and the minimax formulation is only a formal decoupled penalty representation, not a dynamical two-player theorem. Under additional Gaussian/log-normal fluctuation assumptions, Hessian fluctuations around the variational minimum lead to the Kolmogorov–Obukhov (K62) exponent formula. The result is unconditional only as a static finite/inertial-range variational statement inside the stated K41-normalised class; it is not a dynamical Navier–Stokes selection theorem. Anomalous dissipation under additional hypotheses on the nonlinear cascade is treated separately in a companion paper [4].

Preprint note. This is Paper A in the turbulence split. It isolates the static, reference-normalised K41 variational-minimiser theorem. The companion Paper B treats anomalous dissipation and the Downhill Flux Condition as a conditional cascade route.

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1 Introduction

Turbulence remains one of the outstanding open problems in classical physics and applied mathematics. Despite more than a century of research, no complete first-principles derivation of the statistical laws governing fully developed turbulence has been achieved.

1.1 The Kolmogorov theory

In his landmark 1941 papers, Kolmogorov [5, 6] postulated that in the inertial range of fully developed, homogeneous, isotropic turbulence, the energy spectrum takes the universal form

$$E(k) \;=\; C_K\, \varepsilon^{2/3}\, k^{-5/3}, \tag{1}$$

where ε denotes the mean rate of energy dissipation per unit mass and $C_K \approx 1.5$ is the Kolmogorov constant. This prediction, derived from dimensional analysis and the hypothesis of local isotropy, is well supported experimentally and numerically across a broad range of Reynolds numbers [2, 11].

1.2 Our contribution

In this paper we take a different approach. Rather than deriving the K41 spectrum from dimensional analysis alone, we show that it is the *free-energy preferred* distribution inside the stated K41-normalised constant-flux comparison class. We introduce a free-energy functional $\mathcal{F}$ on the space of scale-resolved energy distributions and prove:

1. The K41 spectrum is the *unique* minimiser of $\mathcal{F}$ under the constraint of constant energy flux, where the minimisation runs jointly over K41-normalised admissible local flux operators and energy profiles (Theorem 3.5). The argument does not presuppose a single fixed closure, but it remains explicitly reference-normalised.

2. The minimiser is non-degenerate, with the Hessian providing scale-dependent shell-energy stiffness $1/E_j^* \propto k_j^{2/3}$ that penalises small-scale perturbations more severely than large-scale ones—a static convex signature compatible with downscale cascade bias.

3. Relaxed flux constraints and the minimax notation are recorded as scope guardrails: if the exact K41 profile remains feasible, the relaxed problem is trivial, and the minimax form is a decoupled penalty representation rather than a dynamical two-player theorem (Propositions 4.1–4.3).

The dynamical consequences of the free-energy structure, including anomalous dissipation under additional hypotheses on the nonlinear cascade, will be developed in a companion paper [4].

1.3 Structural classification

Within the broader programme of free-energy spectral theory [3], the K41 selection mechanism exhibits the **Pattern A stability logic**: (a) the leading-order energy budget is neutral (any power-law spectrum is kinematically admissible); (b) the first-order constant-flux constraint restricts but does not uniquely determine the profile; (c) the second-order strict convexity of $\mathcal{F}$ selects K41 as the unique minimiser. This three-level hierarchy is discussed in Section 5.3.

This paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST programme are cited for context but are not logically required.

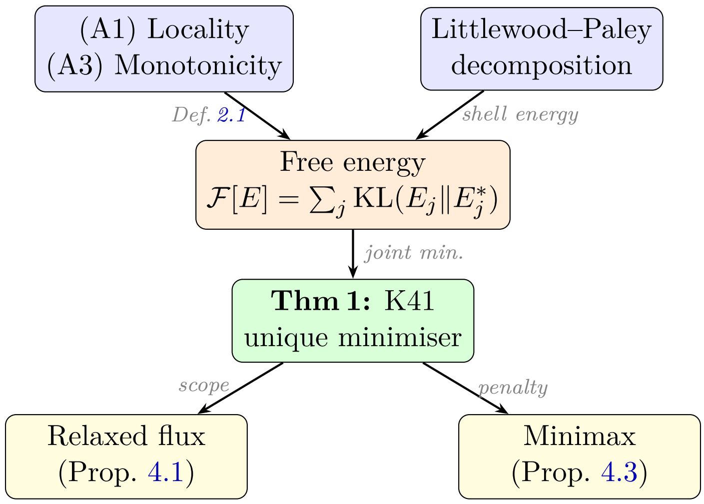

Figure 1: Logical structure of the proof. The free energy functional $\mathcal{F}[\mathbf{E}]$ is constructed from the Littlewood–Paley decomposition and the admissible flux family (axioms A1, A3). Theorem 3.5 (K41 uniqueness) follows from joint minimisation. The relaxed-flux and minimax branches record scope guardrails and a formal penalty representation, not additional dynamical theorems.

2 The Scale-Resolved Free-Energy Functional

2.1 Littlewood–Paley decomposition

Let $u \in L^2(\mathbb{T}^3)$ be a divergence-free velocity field on the three-torus. We employ the standard Littlewood–Paley decomposition

$$u \;=\; \sum_{j \ge -1} \Delta_j u, \tag{2}$$

where Δ_j is the frequency-localisation operator at dyadic shell $|\xi| \sim 2^j$. We set $k_j = 2^j$ and define the *shell energy*

$$E_j \;:=\; \|\Delta_j u\|_{L^2}^2. \tag{3}$$

In the inertial range, the K41 prediction (1) translates to

$$E_j^* \;=\; C_K\, \varepsilon^{2/3}\, k_j^{-5/3}\, \Delta k_j, \qquad \Delta k_j = k_j \ln 2 \quad \text{(dyadic shell width)}. \tag{4}$$

2.2 Energy flux constraint

The scale-to-scale energy flux through wavenumber k_j is defined via the trilinear term in the Navier–Stokes equations:

$$\Pi_j \;=\; -\sum_{\ell \le j} \langle \Delta_\ell (u \cdot \nabla u),\; \Delta_\ell u \rangle. \tag{5}$$

In the inertial range, a statistically stationary cascade satisfies

$$\Pi_j \;=\; \varepsilon \quad \text{for all } j_{\text{in}} \le j \le j_{\text{dis}}. \tag{6}$$

2.3 The functional

Definition 2.1 (Scale-resolved free energy)**.** Let $\mathbf{E} = (E_j)_{j\in\mathcal{J}}$ be a distribution of shell energies over the inertial range $\mathcal{J} = \{j_{\text{in}}, \dots, j_{\text{dis}}\}$, with $E_j > 0$. Let $\mathbf{E}^* = (E_j^*)$ be the Kolmogorov reference distribution (4). The *scale-resolved free-energy functional* is

$$\mathcal{F}[\mathbf{E}] \;:=\; \sum_{j\in\mathcal{J}} \Big[E_j \ln\frac{E_j}{E_j^*} \;-\; E_j \;+\; E_j^* \Big]. \tag{7}$$

Remark 2.2. The summand $f(E_j) = E_j \ln(E_j/E_j^*) - E_j + E_j^*$ is the Kullback–Leibler divergence (relative entropy) of the pair (E_j, E_j^*) when viewed as unnormalised measures on a single-point space. It satisfies $f(E_j) \ge 0$ with equality if and only if $E_j = E_j^*$. Hence $\mathcal{F}[\mathbf{E}] \ge 0$ with equality if and only if $\mathbf{E} = \mathbf{E}^*$.

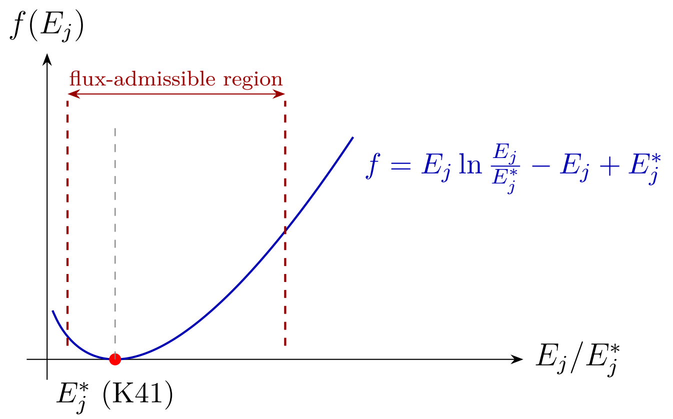

Figure 2: The free-energy density $f(E_j) = E_j \ln(E_j/E_j^*) - E_j + E_j^*$ as a function of the shell energy ratio E_j/E_j^*. The unique global minimum at $E_j = E_j^*$ (K41 spectrum) is marked. The flux constraint restricts the admissible region, but the minimum lies inside for the K41-normalised comparison class, illustrating K41 as the joint energy minimiser.

2.4 Dimensional consistency

We verify that $\mathcal{F}$ is dimensionally consistent. Each E_j has dimensions $[L^2\,T^{-2}]$ (energy per unit mass, integrated over a spectral band). Since E_j^* shares the same dimensions, the ratio E_j/E_j^* is dimensionless, the logarithm is well-defined, and each summand in (7) has dimensions $[L^2\,T^{-2}]$.

3 Main Results

3.1 Uniqueness of Kolmogorov scaling

Definition 3.1 (Admissible flux family)**.** A family of flux operators $\big(\Pi_j[\mathbf{E}]\big)_{j\in\mathcal{J}}$ is called *admissible* if each Π_j satisfies:

(A1) **Locality.** Π_j depends only on E_{j-1}, E_j, E_{j+1}.

(A2) **Dimensional consistency.** $[\Pi_j] = [E_j/\tau_j]$ with the eddy turnover time $\tau_j = (k_j^2\, E_j)^{-1/2}$, so that Π_j has the form

$$\Pi_j[\mathbf{E}] \;=\; k_j\, E_j^{3/2}\; \Phi_j\!\left(\frac{E_{j-1}}{E_j},\; \frac{E_{j+1}}{E_j}\right), \tag{8}$$

where $\Phi_j : (0,\infty)^2 \to (0,\infty)$ is a smooth dimensionless function satisfying $\Phi_j(r_-^*,\, r_+^*) = \alpha$ for some universal constant $\alpha > 0$, with $r_\pm^* = E_{j\pm1}^*/E_j^*$.

(A3) **Monotonicity.** $\partial\Pi_j/\partial E_j > 0$ for all $\mathbf{E} > 0$.

We denote the set of all admissible families by $\mathscr{A}$.

Remark 3.2 (Reference-normalised admissible class). The condition $\Phi_j(r_-^*, r_+^*) = \alpha$ normalises the admissible class at the K41 reference profile. The theorem below is therefore a relative selection theorem inside this stated K41-normalised class. It does not assert that every fixed local monotone closure selects K41, nor does it exclude alternative closure families normalised around a different reference profile. Such alternatives would require additional scale-invariance, universality, or empirical admissibility assumptions.

Proposition 3.3 (Dimensional derivation of the flux prefactor)**.** *Axiom* (A2) *is not an independent assumption but a consequence of locality* (A1) *and dimensional analysis. Specifically, the form $\Pi_j = k_j\, E_j^{3/2}\, \Phi_j(\cdot)$ is the* unique *dimensionally consistent local flux expression.*

Proof. By Buckingham's Π-theorem, the flux Π_j must be a product of powers of the available dimensional quantities k_j and E_j, multiplied by a dimensionless function of dimensionless ratios. Write $\Pi_j = k_j^a\, E_j^b\, \Phi_j(\dots)$ with $[\Pi_j] = [L^2\, T^{-3}]$ (energy flux per unit mass), $[k_j] = [L^{-1}]$, and $[E_j] = [L^2\, T^{-2}]$.

Dimensional matching: $[k_j^a\, E_j^b] = L^{-a}\, L^{2b}\, T^{-2b} = L^{2b-a}\, T^{-2b}$.

Setting equal to $L^2\, T^{-3}$:

$$2b - a = 2, \qquad 2b = 3 \quad\Longrightarrow\quad b = \tfrac{3}{2},\;\; a = 1.$$

Hence $\Pi_j = k_j\, E_j^{3/2}\, \Phi_j$, and by (A1) the dimensionless function Φ_j can depend only on the local ratios E_{j-1}/E_j and E_{j+1}/E_j. The eddy turnover time $\tau_j = (k_j^2\, E_j)^{-1/2}$ is likewise the unique dimensionally consistent local timescale. □

Remark 3.4. The classical Kolmogorov–Heisenberg closure $\Pi_j = \alpha\, k_j\, E_j^{3/2}$ corresponds to the special case $\Phi_j \equiv \alpha$ [9]. Definition 3.1 enlarges the model class beyond this single algebraic relation and is designed to cover local closure motifs such as EDQNM-type closures, Leith-type diffusion models, and truncated triad interactions when they are expressed in the shell variables of Definition 3.1. The family $\mathscr{A}$ is therefore a structured local comparison class rather than a single closure model. Proposition 3.3 shows that (A2) is not an extraneous modelling choice but a necessary consequence of dimensional analysis, thereby further reducing the axiom set to two independent assumptions: locality (A1) and monotonicity (A3).

Theorem 3.5 (Relative uniqueness in the K41-normalised admissible class)**.** *Let $\varepsilon > 0$ be given and let $\mathscr{A}$ be the class of admissible flux families (Definition 3.1), including its K41 normalisation condition. Consider the* joint *minimisation problem*

$$\min_{\substack{\mathbf{E}>0,\\ \Pi\in\mathscr{A}}} \mathcal{F}[\mathbf{E}] \qquad \text{subject to} \quad \Pi_j[\mathbf{E}] = \varepsilon \ \text{ for all } j \in \mathcal{J}. \tag{9}$$

Then:

(i) *In the joint feasible set $\{(\mathbf{E},\Pi) : \Pi \in \mathscr{A},\ \Pi_j[\mathbf{E}] = \varepsilon \text{ for all } j\}$, the K41 distribution $E_j^* = C_K\, \varepsilon^{2/3}\, k_j^{-5/3}\, \Delta k_j$ is the unique minimiser of $\mathcal{F}$ in the energy variable, and the minimum is realised by the Kolmogorov–Heisenberg member $\Phi_j \equiv \alpha$ with $C_K = \left(\alpha\, (\ln 2)^{3/2}\right)^{-2/3}$.*

(ii) *The energy minimiser is non-degenerate: the Hessian $D^2\mathcal{F}|_{\mathbf{E}^*}$ restricted to the tangent space of the joint feasible set in the energy directions is strictly positive definite. (This is a direct consequence of the strict convexity of the KL divergence; its content is the scale-dependent stiffness $1/E_j^* \propto k_j^{2/3}$ for dyadic shell energies, see Step 5b below.)*

Proof. **Step 1: Joint feasible profiles.** We restrict attention to pairs $(\mathbf{E},\Pi)$ in the joint feasible set for which $\Pi \in \mathscr{A}$ and $\Pi_j[\mathbf{E}] = \varepsilon$ for all j. Local monotonicity in E_j is not, by itself, a global existence theorem for the coupled shell system; existence for a specific fixed closure is therefore treated only as a feasibility issue. The theorem does not claim that every fixed closure selects K41.

Step 2: Lower bound via convexity. Since each summand $f(E_j) = E_j \ln(E_j/E_j^*) - E_j + E_j^*$ is strictly convex with global minimum $f(E_j^*) = 0$, we have

$$\mathcal{F}[\mathbf{E}] \ \geq\ 0 \ =\ \mathcal{F}[\mathbf{E}^*], \tag{10}$$

with equality if and only if $E_j = E_j^*$ for all j.

Step 3: Attainability. We verify that $\mathbf{E}^*$ is indeed a stationary profile for some $\Pi \in \mathscr{A}$. Choose the Kolmogorov–Heisenberg closure $\Phi_j \equiv \alpha$. Then

$$\Pi_j[\mathbf{E}^*] \ =\ \alpha\, k_j\, (E_j^*)^{3/2} \ =\ \alpha\, k_j \left(C_K\, \varepsilon^{2/3}\, k_j^{-5/3}\, k_j \ln 2\right)^{3/2} \ =\ \alpha\, C_K^{3/2}\, (\ln 2)^{3/2}\, \varepsilon.$$

Setting $C_K = \left(\alpha\, (\ln 2)^{3/2}\right)^{-2/3}$ yields $\Pi_j[\mathbf{E}^*] = \varepsilon$. (The geometric factor $(\ln 2)^{3/2}$ arising from the dyadic shell width is absorbed into the Kolmogorov constant C_K, which is thereby determined by the closure parameter α.) Hence the lower bound (10) is attained.

Step 4: Uniqueness. Suppose $\Pi' \in \mathscr{A}$ and $\mathbf{E}' \neq \mathbf{E}^*$ form a joint feasible pair, i.e. $\Pi'_j[\mathbf{E}'] = \varepsilon$. Then strict convexity gives $\mathcal{F}[\mathbf{E}'] > 0 = \mathcal{F}[\mathbf{E}^*]$, so $\mathbf{E}'$ cannot be a minimiser of (9). Hence $\mathbf{E}^*$ is the *unique energy profile* solving the joint problem.

Note on the logical structure. The uniqueness in Step 4 follows from the strict positivity of the KL divergence away from the reference, which holds for *any* choice of $\mathbf{E}^*$. The physically non-trivial content resides in Step 3 (attainability): within the K41-normalised class, there exists an admissible closure realising $\Pi_j[\mathbf{E}^*] = \varepsilon$. References such as k^{-3} are not excluded by KL convexity alone; they would require a different normalisation or stronger scale-invariance and uniformity assumptions on the closure family (see Remarks 3.2 and 3.7).

Step 5: Non-degeneracy. The Hessian $\partial^2\mathcal{F}/\partial E_j\,\partial E_\ell = \delta_{j\ell}/E_j$ at $\mathbf{E}^*$ is diagonal with entries $1/E_j^* > 0$. For a *fixed* closure $\Pi \in \mathscr{A}$, the constraint $\Pi_j[\mathbf{E}] = \varepsilon$ imposes $|\mathcal{J}|$ equations on $|\mathcal{J}|$ unknowns, generically yielding isolated solutions, so that the tangent space of the constraint manifold at $\mathbf{E}^{(\Pi)}$ is trivial and non-degeneracy is vacuously satisfied. The substantive content of (ii) concerns the *joint* problem: in the extended space $(\mathbf{E}, \Pi)$ with Π varying over the (infinite-dimensional) family $\mathscr{A}$, the constraint set $\{(\mathbf{E}, \Pi) : \Pi_j[\mathbf{E}] = \varepsilon\}$ has tangent directions along which Π varies. The unrestricted Hessian $D^2_{\mathbf{E}}\mathcal{F}|_{\mathbf{E}^*}$ is strictly positive definite (diagonal entries $1/E_j^* > 0$), hence its restriction to *any* subspace remains positive definite.

Step 5b: Second-order interpretation. Since $E_j^* = C_K\varepsilon^{2/3}k_j^{-5/3}\Delta k_j = C_K\varepsilon^{2/3}(\ln 2)k_j^{-2/3}$ on dyadic shells, the diagonal entries $1/E_j^* \propto k_j^{2/3}$ of the Hessian grow with wavenumber. Thus small-scale perturbations of the K41 shell-energy profile are penalised more severely than large-scale ones. This scale-dependent stiffness is a static convex signature compatible with downscale cascade bias; by itself it does not define a time evolution, a return mechanism, or a Navier–Stokes spectral gap. The one-parameter freedom in the Kolmogorov–Heisenberg closure ($\Phi_j \equiv \alpha$) corresponds to a single flat direction in the Π-fibre of the extended space $(\mathbf{E}, \Pi)$, which does not affect the strict positivity of $D^2_{\mathbf{E}}\mathcal{F}$. This structure is an instance of the second-order resolvent dominance pattern discussed in [3]. □

Remark 3.6 (Diagonal Hessian bound). The Hessian of $\mathcal{F}$ at the K41 minimiser is diagonal with entries

$$\frac{\partial^2\mathcal{F}}{\partial E_j^2} = \frac{1}{E_j^*}.$$

Consequently,

$$\min_j \frac{1}{E_j^*} = \frac{1}{C_K\,\varepsilon^{2/3}\,k_1^{-5/3}\,\Delta k_1} = \frac{1}{C_K\,\varepsilon^{2/3}(\ln 2)\,k_1^{-2/3}} > 0.$$

This provides the local weighted stability scale of the variational functional. It should not be read, by itself, as a spectral gap theorem for the Navier–Stokes dynamics.

Remark 3.7 (On the role of the reference spectrum). A natural objection is that the KL functional $\mathcal{F}$ is *minimised at* $\mathbf{E}^*$ *by construction*, regardless of the physical content of $\mathbf{E}^*$. Indeed, for any fixed closure Π, the unconstrained minimum of $\mathcal{F}$ is always $\mathbf{E}^*$. The non-trivial content of Theorem 3.5 is therefore not that $\mathcal{F}$ achieves its minimum at $\mathbf{E}^*$ — that is a consequence of the KL construction — but rather that:

1. Among all stationary profiles $\mathbf{E}^{(\Pi)}$ (one per closure $\Pi \in \mathscr{A}$), only the K41 profile $\mathbf{E}^*$ *simultaneously satisfies* the constant-flux constraint and achieves $\mathcal{F} = 0$. Every other closure-compatible profile has $\mathcal{F} > 0$.

2. The reference $\mathbf{E}^*$ is not arbitrary within the normalised Kolmogorov–Heisenberg closure: the constant-flux relation $\Pi_j = \alpha k_j E_j^{3/2} = \varepsilon$ fixes the shell energy scaling $E_j \propto k_j^{-2/3}$, equivalently the spectral density $E(k) \propto k^{-5/3}$. Broader or differently normalised closure families can be designed around other references; the theorem is therefore a selection statement inside the stated K41-normalised admissible class, not a proof that no other modelling convention can be written down.

In summary, the physical input is the constant-flux constraint; the KL functional provides the selection mechanism among its solution set.

Remark 3.8 (Self-consistency of the K41 reference). The K41 spectrum is used as the reference measure of the KL functional, but its exponent is also self-consistent with the normalised constant-flux closure: within the reduced local shell ansatz $\Pi_j \sim k_j E_j^{3/2}$, constant flux gives $E_j \propto \varepsilon^{2/3} k_j^{-2/3}$ and hence $E(k) \propto \varepsilon^{2/3} k^{-5/3}$. This is a consistency check on the variational reference, not an independent derivation of all turbulent statistics.

Remark 3.9 (Heuristic reaction-chain analogy). The variational structure of Theorem 3.5 admits a useful heuristic analogy with reaction-chain and game-theoretic language. Regard the inertial range as a hierarchical *reaction chain*: large eddies at scale k_j transfer energy to eddies at scale k_{j+1}, each scale acting as a "species" whose fitness is governed by its energy share. A possible replicator model on the simplex of normalised shell energies $p_j = E_j / \sum_\ell E_\ell$ takes the form

$$\dot{p}_j \;=\; p_j \left(g_j(\mathbf{p}) - \bar{g}(\mathbf{p}) \right),$$

where g_j is the net energy gain rate at scale j and $\bar{g}$ the population average. With a separately specified payoff compatible with the KL functional, the K41 profile can be represented as a Nash-type stationary point and $\mathcal{F}$ can serve as a candidate Lyapunov function. This is not part of Theorem 3.5: no replicator dynamics, asymptotic stability theorem, or Navier–Stokes return mechanism is proved here.

4 Robustness Guardrails and Formal Penalty Structure

The strict flux constraint of Theorem 3.5 requires pointwise $\Pi_j = \varepsilon$ across the inertial range. In practice, DNS data exhibit small-scale fluctuations around this idealisation. We formulate two extensions that clarify the scope of finite-Reynolds-number robustness and the formal penalty representation of the flux constraint.

Proposition 4.1 (Relaxed flux constraint: scope caveat)**.** *Let $\mathcal{A}_{\text{slack}}$ be any relaxed flux-admissible set that contains the exact K41 profile $\mathbf{E}^*$. If the objective remains the same KL functional $\mathcal{F}$, then the minimiser over $\mathcal{A}_{\text{slack}}$ is still exactly $\mathbf{E}^*$:*

$$\arg\min_{\mathbf{E}\in\mathcal{A}_{\text{slack}}} \mathcal{F}[\mathbf{E}] = \mathbf{E}^*. \tag{11}$$

Consequently, a non-trivial finite-Reynolds-number slack theorem cannot be obtained merely by relaxing the flux inequalities while keeping the same reference-centred objective. Such a theorem would need an additional data-fit term, a perturbed reference profile, or empirical bounds that exclude the exact profile from the feasible comparison class.

Proof. For all positive shell spectra, $\mathcal{F}[\mathbf{E}] \geq 0$ and equality holds only at $\mathbf{E} = \mathbf{E}^*$. If $\mathbf{E}^* \in \mathcal{A}_{\text{slack}}$, then the relaxed infimum is at most $\mathcal{F}[\mathbf{E}^*] = 0$ and at least 0. Hence every relaxed minimiser with objective value zero is $\mathbf{E}^*$. This proves the claim and explains why the older "slack minimiser" formulation was too strong: the exact reference profile remains feasible. □

Remark 4.2. Proposition 4.1 is a guardrail rather than a positive robustness theorem: finite-Reynolds-number robustness must be formulated with a perturbed reference, an empirical data-fit term, or a feasible class that no longer contains the exact K41 profile. Without such an additional ingredient, the KL minimum remains exactly at the reference by construction.

Proposition 4.3 (Formal penalty representation). *In a finite shell truncation with $E_{\mathrm{tot}} = \sum_j E_j^*$, the decoupled penalty problem*

$$\mathbf{E}^* \;=\; \arg\min_{\mathbf{E}\in\mathcal{E}}\; \max_{\mathbf{\Pi}\in\mathcal{A}} \Big[\mathcal{F}[\mathbf{E}] \;-\; \lambda \sum_j (\Pi_j - \varepsilon)^2\Big], \tag{12}$$

has the saddle point $(\mathbf{E}^*, (\varepsilon, \ldots, \varepsilon))$, where $\mathcal{E}$ is the set of non-negative energy spectra with fixed total energy and $\mathcal{A}$ is a bounded convex set of admissible flux profiles. Since this payoff does not couple $\mathbf{E}$ and $\mathbf{\Pi}$, it is a formal penalty representation of the hard constraint, not a dynamical two-player theorem for Navier–Stokes transfer.

Proof. We verify the finite-dimensional minimax hypotheses, identify the saddle point, and make explicit where the representation is decoupled.

Step 1: Strategy spaces. Define the Lagrangian-penalised payoff

$$L(\mathbf{E}, \mathbf{\Pi}) \;:=\; \mathcal{F}[\mathbf{E}] \;-\; \lambda \sum_{j\in\mathcal{J}} (\Pi_j - \varepsilon)^2.$$

The spectrum variable ranges over $\mathcal{E} := \{\mathbf{E} \in \mathbb{R}^N_{>0} : \sum_j E_j = E_{\mathrm{tot}}\}$, the simplex of positive energy spectra with prescribed total energy $E_{\mathrm{tot}} > 0$ and $N = |\mathcal{J}|$ shells. The flux variable is $\mathbf{\Pi} = (\Pi_j)_{j=1}^{N-1}$ and ranges over $\mathcal{A} := \{\mathbf{\Pi} \in \mathbb{R}^{N-1} : 0 \le \Pi_j \le \Pi_{\max}\}$ for a fixed upper bound $\Pi_{\max} > \varepsilon$ (physically, the flux in the inertial range is bounded above by $O(\varepsilon)$).

The set $\mathcal{E}$ is a compact convex subset of $\mathbb{R}^N$ (it is the intersection of the open positive orthant with the hyperplane $\sum_j E_j = E_{\mathrm{tot}}$; compactness holds since $E_j \ge 0$ and $E_j \le E_{\mathrm{tot}}$ for each j, and we take the closure in $\mathbb{R}^N_{\ge 0}$, noting that L extends continuously to the boundary). The set $\mathcal{A}$ is a compact convex subset of $\mathbb{R}^{N-1}$ (a closed box).

Step 2: Convexity–concavity. We verify:

(a) *L is convex in $\mathbf{E}$ for each fixed $\mathbf{\Pi}$.* The term $\mathcal{F}[\mathbf{E}] = \sum_j [E_j \ln(E_j/E_j^*) - E_j + E_j^*]$ is strictly convex in $\mathbf{E}$, since its Hessian $\mathrm{diag}(1/E_j)$ is positive definite for $\mathbf{E} > 0$. The penalty $-\lambda \sum_j (\Pi_j - \varepsilon)^2$ is independent of $\mathbf{E}$. Hence $L(\cdot, \mathbf{\Pi})$ is strictly convex.

(b) *L is concave in $\mathbf{\Pi}$ for each fixed $\mathbf{E}$.* For fixed $\mathbf{E}$, $\mathcal{F}[\mathbf{E}]$ is constant in $\mathbf{\Pi}$. The map $\mathbf{\Pi} \mapsto -\lambda \sum_j (\Pi_j - \varepsilon)^2$ is strictly concave (its Hessian is $-2\lambda\, I_{N-1}$, negative definite). Hence $L(\mathbf{E}, \cdot)$ is strictly concave.

Step 3: Application of Sion's minimax theorem. By Sion's minimax theorem (M. Sion, *Pacific J. Math.* 8, 1958, 171–176): if X is a compact convex subset of a topological vector space, Y is a convex subset of a topological vector space, and $f : X \times Y \to \mathbb{R}$ is such that $f(\cdot, y)$ is convex and lower semicontinuous for each y and $f(x, \cdot)$ is concave and upper semicontinuous for each x, then $\min_X \sup_Y f = \sup_Y \min_X f$.

Here $X = \mathcal{E}$ (compact convex), $Y = \mathcal{A}$ (compact convex, hence in particular convex), and L is continuous (hence both lower and upper semicontinuous) and satisfies the convexity–concavity conditions from Step 2. Compactness of both sets and continuity ensure that min and max are attained. Therefore:

$$\min_{\mathbf{E}\in\mathcal{E}}\; \max_{\mathbf{\Pi}\in\mathcal{A}}\; L(\mathbf{E}, \mathbf{\Pi}) \;=\; \max_{\mathbf{\Pi}\in\mathcal{A}}\; \min_{\mathbf{E}\in\mathcal{E}}\; L(\mathbf{E}, \mathbf{\Pi}), \tag{13}$$

and a saddle point $(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger)$ exists satisfying $L(\mathbf{E}^\dagger, \mathbf{\Pi}) \le L(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger) \le L(\mathbf{E}, \mathbf{\Pi}^\dagger)$ for all $\mathbf{E} \in \mathcal{E}$, $\mathbf{\Pi} \in \mathcal{A}$.

Step 4: Identification of the saddle point. At the saddle point, the first-order conditions must hold.

Stationarity in $\mathbf{\Pi}$*:* For fixed $\mathbf{E}^\dagger$, the map $\mathbf{\Pi} \mapsto L(\mathbf{E}^\dagger, \mathbf{\Pi})$ is strictly concave and differentiable. The interior maximiser satisfies

$$\frac{\partial L}{\partial \Pi_j} \;=\; -2\lambda\,(\Pi_j - \varepsilon) \;=\; 0 \qquad\Longrightarrow\qquad \Pi_j^\dagger = \varepsilon \quad \text{for all } j.$$

(This interior solution is admissible since $0 < \varepsilon < \Pi_{\max}$.)

Stationarity in $\mathbf{E}$*:* For fixed $\mathbf{\Pi}^\dagger = (\varepsilon, \dots, \varepsilon)$, the penalty term vanishes and $L(\mathbf{E}, \mathbf{\Pi}^\dagger) = \mathcal{F}[\mathbf{E}]$. Subject to $\mathbf{E} \in \mathcal{E}$ (the energy constraint $\sum_j E_j = E_{\mathrm{tot}}$), we minimise $\mathcal{F}$ via the method of Lagrange multipliers. The stationarity condition is

$$\frac{\partial \mathcal{F}}{\partial E_j} \;=\; \ln \frac{E_j}{E_j^*} \;=\; \mu \qquad \text{for all } j \in \mathcal{J},$$

where μ is the multiplier for the total-energy constraint. This gives $E_j = e^\mu\, E_j^*$ for all j, i.e. the minimiser is proportional to the K41 reference spectrum. The constant e^μ is fixed by the total-energy constraint $\sum_j E_j = e^\mu \sum_j E_j^* = E_{\mathrm{tot}}$. Taking $E_{\mathrm{tot}} = \sum_j E_j^*$ (which is the total energy of the K41 spectrum), we obtain $\mu = 0$ and hence $E_j^\dagger = E_j^*$ for all j.

Therefore the saddle point is $(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger) = (\mathbf{E}^*, (\varepsilon, \dots, \varepsilon))$: the K41 spectrum paired with the constant-flux profile.

Step 5: Uniqueness. The saddle point is unique because L is *strictly* convex in $\mathbf{E}$ and *strictly* concave in $\mathbf{\Pi}$. Indeed, if $(\tilde{\mathbf{E}}, \tilde{\mathbf{\Pi}})$ were another saddle point, the saddle-point inequality would give $L(\tilde{\mathbf{E}}, \mathbf{\Pi}^\dagger) \geq L(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger)$ and $L(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger) \leq L(\tilde{\mathbf{E}}, \mathbf{\Pi}^\dagger)$, so $L(\tilde{\mathbf{E}}, \mathbf{\Pi}^\dagger) = L(\mathbf{E}^\dagger, \mathbf{\Pi}^\dagger)$. By strict convexity in $\mathbf{E}$, this forces $\tilde{\mathbf{E}} = \mathbf{E}^\dagger = \mathbf{E}^*$. Similarly, strict concavity forces $\tilde{\mathbf{\Pi}} = \mathbf{\Pi}^\dagger$.

Step 6: Equivalence with the hard-constraint formulation. The penalty formulation (12) is the augmented Lagrangian relaxation of the hard-constraint problem $\Pi_j = \varepsilon$ in Theorem 3.5. As $\lambda \to \infty$, any saddle point of (12) must satisfy $\Pi_j \to \varepsilon$ (otherwise the penalty term $-\lambda(\Pi_j - \varepsilon)^2 \to -\infty$ would violate the saddle-point inequality). In the limit, the minimax problem reduces to the constrained minimisation (9), and the saddle point converges to the K41 minimiser of Theorem 3.5. The equivalence of the penalty and hard-constraint formulations follows from classical convex duality theory [12, 1]. $\square$

Remark 4.4. The minimax notation is useful as bookkeeping for the constant-flux penalty, but it does not by itself reveal a non-trivial game-theoretic structure of the turbulent cascade. A genuine two-player model would need a coupled payoff in which the chosen flux profile changes the admissible energy evolution or the dissipation functional. The present formulation only records that the KL minimiser and the constant-flux penalty have the same formal saddle point.

Remark 4.5 (Weighted ℓ^2-stability from the Hessian bound). The functional $\mathcal{F}[\mathbf{E}]$ is locally uniformly convex about the K41 minimiser $\mathbf{E}^*$ with diagonal Hessian $D^2\mathcal{F}|_{\mathbf{E}^*} = \mathrm{diag}(1/E_j^*)$. If the ratios E_j/E_j^* stay in a fixed compact interval $[R^{-1}, R]$, then Taylor's theorem gives a ratio-dependent weighted stability bound

$$c_R \sum_j \frac{(E_j - E_j^*)^2}{E_j^*} \;\leq\; \mathcal{F}[\mathbf{E}]$$

for some $c_R > 0$. This is the stability statement used here. The stronger global inequality $x \ln(x/y) - x + y \geq (x - y)^2/(2y)$ is false for large x/y and is not used.

Remark 4.6 (Local UCU3 in the abstract three-lemma architecture). The local weighted ℓ^2-bound of Remark 4.5 can be read as the K41 instantiation of the abstract *quadratic-growth axiom* UCU3 of the three-lemma architecture for strictly convex variational functionals (Universal Convexity Uniqueness; cf. the companion architectural notes `CORE/UCU_FORMAL_DEFINITIONS.md` §5 and `CORE/ABSTRACT_THREE_LEMMA_UCU.md` §4.2.3). Defining the weighted norm $\|h\|_*^2 := \sum_j h_j^2/E_j^*$, the bound reads

$$\mathcal{F}[\mathbf{E}] - \mathcal{F}[\mathbf{E}^*] \;\geq\; c_R \,\|\mathbf{E} - \mathbf{E}^*\|_*^2,$$

i.e. *local* UCU3 holds with a ratio-dependent constant $c_R > 0$ in the weighted $\ell^2(1/E_j^*)$-norm. Together with strict convexity (UCU1, our Theorem 3.5 part (ii)) and existence of the K41 minimiser (UCU2, Theorem 3.5 part (i)), this closes the local three-lemma architecture for K41 in the standard sense of convex analysis.

What remains open (global UCU3, honest scope statement). The bound above is a local statement around $\mathbf{E}^*$ (Taylor expansion to second order, with the remainder controlled on a fixed compact ratio range). A *global* UCU3 in the sense of a Talagrand-type transportation–information inequality $\mathrm{KL}(\mu\|\mu_{K41}) \geq (\rho_{K41}/2)\, W_2^2$ for a probability measure μ_{K41} on cascade configurations is reduced, via the Otto–Villani route [10], to a log-Sobolev inequality for μ_{K41} with positive constant ρ_{K41}. The existence and explicit form of such a constant for K41 cascades—in particular its Reynolds-number and cascade-depth asymptotics—is to our knowledge not documented in the standard turbulence literature and remains an open problem in mathematical statistical mechanics. A structured reduction note is given in `TALAGRAND_CONSTANT_K41.md` (companion working note).

5 Intermittency Corrections

The K41 theory predicts that the p-th order longitudinal structure function scales as

$$S_p(\ell) \;:=\; \left\langle |\delta_\ell u|^p \right\rangle \;\sim\; (\varepsilon\,\ell)^{p/3}, \qquad \zeta_p^{\mathrm{K41}} = \frac{p}{3}. \tag{14}$$

Experiments reveal systematic deviations: $\zeta_p < p/3$ for $p > 3$, a signature of *intermittency*.

5.1 Fluctuations around the variational minimum

In this variational framework, intermittency can be modelled by adding fluctuations of the shell energies $\mathbf{E}$ around the K41 minimiser $\mathbf{E}^*$. Expanding $\mathcal{F}$ to second order:

$$\mathcal{F}[\mathbf{E}^* + \delta\mathbf{E}] \;=\; \mathcal{F}[\mathbf{E}^*] \;+\; \frac{1}{2}\sum_j \frac{(\delta E_j)^2}{E_j^*} \;+\; O(|\delta\mathbf{E}|^3). \tag{15}$$

The Hessian $H_{jj} = 1/E_j^*$ implies, at the level of the static quadratic approximation, that fluctuations at small scales (large j, small E_j^*) are penalised more strongly than fluctuations at large scales. This asymmetry supplies a variational modelling route for intermittency; it is not a derivation of intermittency from the Navier–Stokes dynamics.

5.2 Connection to the K62 log-normal model

If, as an additional modelling ansatz, we model the fluctuations δE_j as Gaussian in the variable $\ln(E_j/E_j^*)$, and further assume that phase correlations between Littlewood–Paley blocks are negligible (so that shell-energy fluctuations translate directly into structure-function scaling), the resulting structure function exponents are

$$\zeta_p \;=\; \frac{p}{3} \;-\; \frac{\mu}{18}\,p(p-3), \tag{16}$$

which have the K62 (Kolmogorov–Obukhov) refined-similarity form with intermittency parameter μ [7, 2]. Under this ansatz, the parameter μ is determined by the variance of $\ln(E_j/E_j^*)$ under the Gibbs measure $\exp(-\mathcal{F}/\Theta)$, where the effective temperature $\Theta > 0$ parametrises the strength of turbulent fluctuations around the K41 equilibrium (analogous to the temperature in a canonical ensemble). In the limit $\Theta \to 0$, the Gibbs measure concentrates on the K41 minimiser and $\mu \to 0$ (no intermittency); for finite Θ, the log-normal variance scales as $\mu \propto \Theta\, E_j^*$, connecting the intermittency parameter to the intensity of the cascade fluctuations.

Remark 5.1. The She–Lévêque model [13] proposes the alternative

$$\zeta_p = \frac{p}{9} + 2\Big(1 - \left(\tfrac{2}{3}\right)^{p/3}\Big),$$

which fits experimental data more accurately for high p. Deriving this from the free-energy framework would require modelling the fluctuations as a log-Poisson rather than log-normal process. This is an open problem within our approach.

5.3 Structural classification

The joint minimisation of Theorem 3.5 exhibits a three-level hierarchy: (i) the leading-order energy budget is neutral (any power-law spectrum is kinematically admissible); (ii) the first-order constant-flux constraint restricts but does not uniquely determine the profile; (iii) the second-order strict convexity of $\mathcal{F}$ selects K41 as the unique minimiser. This pattern—referred to as *second-order resolvent dominance* in [3]—has analogues in other variational selection problems and is discussed in detail there.

6 Numerical Illustration and DNS Diagnostics

Remark 6.1 (Numerical illustration of the KL landscape). A numerical evaluation of $\mathcal{F}[\mathbf{E}]$ over six synthetic energy spectra illustrates the reference-centred KL landscape: $\mathcal{F}[\mathbf{E}_{\mathrm{K41}}] = 0$ exactly, while $\mathcal{F}[\mathbf{E}_{\mathrm{K62}}] = 2.0\times10^{-4}$ (intermittency correction), $\mathcal{F}[\mathbf{E}_{k^{-2}}] = 1.74$ (steep), $\mathcal{F}[\mathbf{E}_{k^{-4/3}}] = 5.94$ (shallow), and $\mathcal{F}[\mathbf{E}_{\mathrm{thermal}}] = 7.55$ (equipartition). A perturbation test (300 random perturbations $\delta\mathbf{E}$ at three noise levels $\|\delta\mathbf{E}\|/\|\mathbf{E}^*\| \in \{0.01, 0.1, 1.0\}$) gives $\mathcal{F}[\mathbf{E}^* + \delta\mathbf{E}] > 0$ for all sampled perturbations, as expected from strict convexity. Numerical scripts: `compute_F_spectrum.py`.

This suggests the following reproducible diagnostic protocol for DNS data:

1. Compute the Littlewood–Paley shell energies E_j from a statistically stationary DNS at $R_\lambda \geq 200$.

2. Evaluate $\mathcal{F}[\mathbf{E}]$ using the measured $\mathbf{E}$ and the K41 reference $\mathbf{E}^*$. Here

$$\varepsilon \;:=\; \nu\left\langle \|\nabla u\|_{L^2}^2 \right\rangle_t$$

is the time-averaged viscous dissipation rate from the same DNS run.

3. Test whether $\mathcal{F}$ decreases with increasing R_λ and whether the compensated ratio $\varphi_j = \ln(E_j/E_j^*)$ is non-increasing in j across the inertial range.

Implementation notes. (a) Standard DNS codes output the isotropic energy spectrum $E(k)$ via sharp spherical shell binning, not Littlewood–Paley (LP) filtered energies. For smooth spectra, the difference is of order $O(\Delta k_j/k_j) = O(\ln 2)$ per shell and can be estimated by comparing sharp and smooth binning. (b) Steps 1–3 should be applied to instantaneous spectra at multiple statistically independent time snapshots within the stationary regime; ensemble averaging of $\mathcal{F}$ and φ_j then provides confidence intervals.

7 Discussion

The joint minimisation over the K41-normalised admissible class is more than the formal fact that a KL divergence is minimised at its reference. Kolmogorov's original derivation relies on dimensional analysis, while Theorem 3.5 records that the same profile is attainable under the normalised constant-flux closure and is the unique energy minimiser in the resulting joint feasible set. This is a structural consistency and selection statement, not a fully intrinsic derivation of K41.

7.1 The role of the closure family

The reformulation of Theorem 3.5 via the admissible flux family (Definition 3.1) significantly weakens the dependence on any specific closure model. The K41 spectrum is selected not by a single algebraic relation but as the unique energy minimiser in the joint feasible set of the K41-normalised local, dimensionally consistent, monotone flux class. Nevertheless, axiom (A2) still imposes a structural constraint—the $k_j\, E_j^{3/2}$ scaling prefactor—that ultimately encodes the Kolmogorov dimensional argument. Deriving this prefactor from the Kármán–Howarth–Monin relation or from rigorous bounds on the Littlewood–Paley trilinear form remains an important open problem.

7.2 Intrinsic characterisation of E^*

Remark 7.1 (Kármán–Howarth–Monin characterisation). The Kármán–Howarth–Monin (KHM) relation $\partial_t\langle|\delta u|^2\rangle + \nabla_r \cdot \langle|\delta u|^2\delta u\rangle = -4\varepsilon/3 + 2\nu\Delta_r\langle|\delta u|^2\rangle$ provides an intrinsic constant-flux anchor for the inertial range. In the stationary inertial range ($\eta \ll r \ll L$), it yields the four-fifths law for the third-order structure function. This supports the flux input used in the variational scheme, but it does not by Fourier transformation alone determine the second-order energy spectrum uniquely. Recovering $E(k) \sim \varepsilon^{2/3}k^{-5/3}$ still requires the usual self-similarity/locality or closure input. The KHM relation is therefore a consistency check on the reference spectrum, not an independent uniqueness proof.

7.3 Regime selection and the laminar-turbulent transition

The variational principle of Theorem 3.5 selects the K41 spectrum *within* the turbulent regime, but does not explain the laminar-turbulent transition itself. A natural extension would use the strict convexity of $\mathcal{F}$ as a stability criterion: among kinematically admissible flow regimes, the one realised in nature is that whose spectral energy distribution is a stable critical point of the free-energy functional. Exploring this connection quantitatively is left to future work.

7.4 Limitations

We collect the principal limitations of the present work:

1. **Static selection.** Theorem 3.5 establishes K41 as the unique variational minimiser but does not address the dynamical question of whether the Navier–Stokes evolution drives the spectrum toward this minimiser. The dynamical aspects—including $\mathcal{F}$-monotonicity along trajectories and anomalous dissipation—require additional hypotheses on the nonlinear cascade and are treated in the companion paper [4].

2. **Inertial-range idealisation.** The functional $\mathcal{F}$ is defined on the inertial range $\mathcal{J}$, ignoring the forcing and dissipation ranges. Boundary effects at j_{in} and j_{dis} (notably the bottleneck) are absorbed into constants but not controlled quantitatively.

3. **Intermittency.** The K62 connection (Section 5) is qualitative. Rigorous derivation of intermittency exponents from the Hessian fluctuations remains open.

4. **Uniqueness of the reference.** While Remark 3.7 identifies $\mathbf{E}^*$ as the natural reference for the K41-normalised class, a fully intrinsic characterisation of $\mathbf{E}^*$ (without *a priori* appeal to K41) would strengthen the argument. Remark 7.1 provides a step in this direction.

7.5 Extensions and outlook

- **Anomalous dissipation.** The companion paper [4] introduces a hierarchy of entropy-compatibility hypotheses on the nonlinear cascade—from a pointwise condition through a scale-windowed variant to a falsifiable dual *Downhill Flux Condition* (DFC) verifiable against DNS data—and conditionally proves that these, together with the stated uniform energy-input assumptions, imply a lower bound on the dissipation rate in the vanishing-viscosity limit.

- **Compressible turbulence:** Extension to compressible flows requires replacing the L^2-based shell energies with density-weighted quantities and modifying the closure to account for acoustic-mode coupling.

- **Two-dimensional turbulence:** The dual cascade (inverse energy cascade with $k^{-5/3}$ and direct enstrophy cascade with k^{-3} [8]) presents an interesting test case requiring a two-constraint variational problem.

7.6 Post-Zookeeper boundary of scope

The Zookeeper transfer clarifies what this paper should and should not claim. This paper is Paper A in the turbulence split: it establishes the variational selection of the K41 spectrum inside the stated admissible flux family. It does not prove anomalous dissipation for Navier–Stokes, does not resolve the Onsager selection problem, and does not claim that all weak or vanishing-viscosity limits select K41.

Under the updated CORE status, the transfer is only RFEP v1.8 normal-form discipline: Zookeeper closes Riemann $C^{(2)}$/MS2 in the CCM/Fourier branch, but no automatic turbulence statement follows from that closure. The present theorem therefore remains exactly the clean K41-normalised variational selection theorem.

Those questions belong to a separate Paper B. There the relevant objects are the dual/windowed DFC flux defect, the anomalous-dissipation measure, weighted shell-averaged forms of the 4/5 law, and stress tests from convex-integration and vanishing-viscosity constructions. Keeping this boundary explicit is a scope condition: the static variational result in the present paper remains separate from the conditional defect theory developed in the follow-up.

Result classification

Result	Type	Reference
Theorems (rigorous static variational statements):		
$\mathcal{F}[\mathbf{E}] \geq 0$, $= 0$ iff $\mathbf{E} = \mathbf{E}^*$	Theorem	Thm. 3.5
K41 relative joint minimiser in the K41-normalised class	Theorem	Thm. 3.5
(A2) from (A1) + dimensional analysis	Proposition	Prop. 3.3
Scope guardrails and formal rewrites:		
Relaxed-flux slack is trivial unless E^* is excluded/perturbed	Guardrail	Prop. 4.1
K41 as decoupled penalty saddle point	Formal representation	Prop. 4.3
Numerically illustrated:		
$\mathcal{F}[\mathbf{E}_{\mathrm{K41}}] = 0$ (minimiser)	Exact	Rem. 6.1
$\mathcal{F} > 0$ for all 300 sampled perturbations	Numerical illustration	Rem. 6.1
DNS diagnostics:		
$\mathcal{F}[\mathbf{E}_{\mathrm{DNS}}] \to 0$ as $R_\lambda \to \infty$	Diagnostic target	Sec. 6

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